ISWE406L - AGILE DEVELOPMENT PROCESS AND DEVOPS

COURSE PROJECT

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Slot: C1+TC1

69. Meeting Scheduling System

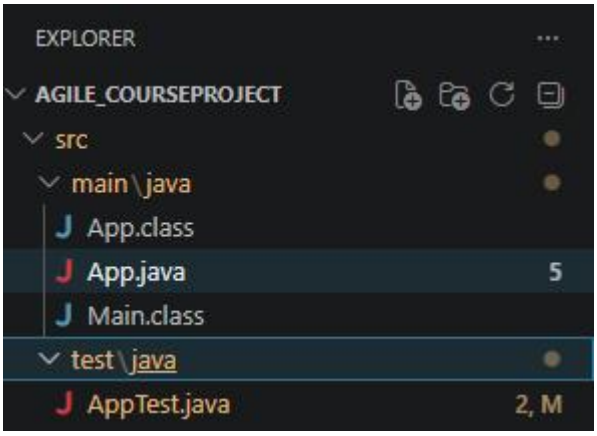
Problem Statement:
Ensure meetings are scheduled without conflicts among participants.

Tasks:

- Develop scheduling logic
- Write JUnit tests for:
  - Time conflict detection
  - Invalid scheduling scenarios
- CI/CD pipeline for automated deployment
- Deploy on Kubernetes

Real-world relevance: Corporate scheduling tools

Create file directory:



App.java

```
import java.time.LocalDateTime;
import java.util.ArrayList;
import java.util.List;

class Meeting {
    String title;
    LocalDateTime start;
    LocalDateTime end;
    public Meeting(String title, LocalDateTime start, LocalDateTime end) {
        if (start.isAfter(end) || start.isEqual(end)) {
            throw new IllegalArgumentException("Invalid time range");
        }
        this.title = title;
        this.start = start;
        this.end = end;
    }
}
```

```java
public class App {
    private List<Meeting> meetings = new ArrayList<>();
    // Schedule meeting
    public boolean scheduleMeeting(Meeting newMeeting) {
        for (Meeting m : meetings) {
            if (isConflict(m, newMeeting)) {
                return false;
            }
        }

        meetings.add(newMeeting);
        return true;
    }

    // Conflict detection
    public boolean isConflict(Meeting m1, Meeting m2) {
        return m1.start.isBefore(m2.end) && m2.start.isBefore(m1.end);
    }
    public List<Meeting> getMeetings() {
        return meetings;
    }
}
```

AppTest.java

```java
import java.time.LocalDateTime;

import static org.junit.jupiter.api.Assertions.assertFalse;
import static org.junit.jupiter.api.Assertions.assertThrows;
import static org.junit.jupiter.api.Assertions.assertTrue;
import org.junit.jupiter.api.Test;

public class AppTest {
    @Test
    public void testNoConflict() {
        App app = new App();

        Meeting m1 = new Meeting("M1",
                LocalDateTime.of(2026, 4, 8, 10, 0),
                LocalDateTime.of(2026, 4, 8, 11, 0));

        Meeting m2 = new Meeting("M2",
                LocalDateTime.of(2026, 4, 8, 11, 0),
                LocalDateTime.of(2026, 4, 8, 12, 0));

        assertTrue(app.scheduleMeeting(m1));
        assertTrue(app.scheduleMeeting(m2));
    }
```

```java
    @Test
    public void testConflict() {
        App app = new App();

        Meeting m1 = new Meeting("M1",
                LocalDateTime.of(2026, 4, 8, 10, 0),
                LocalDateTime.of(2026, 4, 8, 11, 0));

        Meeting m2 = new Meeting("M2",
                LocalDateTime.of(2026, 4, 8, 10, 30),
                LocalDateTime.of(2026, 4, 8, 11, 30));

        app.scheduleMeeting(m1);
        assertFalse(app.scheduleMeeting(m2));
    }

    @Test
    public void testInvalidTime() {
        assertThrows(IllegalArgumentException.class, () -> {
            new Meeting("Invalid",
                    LocalDateTime.of(2026, 4, 8, 12, 0),
                    LocalDateTime.of(2026, 4, 8, 11,
                     0));
        });
    }

    @Test
    public void testExactOverlap() {
        App app = new App();

        Meeting m1 = new Meeting("M1",
                LocalDateTime.of(2026, 4, 8, 10, 0),
                LocalDateTime.of(2026, 4, 8, 11, 0));

        Meeting m2 = new Meeting("M2",
                LocalDateTime.of(2026, 4, 8, 10, 0),
                LocalDateTime.of(2026, 4, 8, 11, 0));

        app.scheduleMeeting(m1);
        assertFalse(app.scheduleMeeting(m2));
    }
}
```

Create Pom.xml file

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.demo</groupId>
<artifactId>test-java-app</artifactId>
<version>1.0</version>
<packaging>jar</packaging>
<name>test-java-app</name>
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
<junit.version>5.10.1</junit.version>
</properties>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<version>${junit.version}</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>3.2.5</version>
<configuration>
<useModulePath>false</useModulePath>
</configuration>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.4.1</version>
<configuration>
<archive>
<manifest>
<mainClass>App</mainClass>
</manifest>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Create Dockerfile

Dockerfile

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Create deployment.yaml file

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:
        app: java-app
    spec:
      containers:
      - name: java-app
        image: madhumitha3005/java-app:latest ---
apiVersion: v1
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30008
```

mvn clean test

```
PROBLEMS 7   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   powershell

PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\NARESH AS\Desktop\Agile_CourseProject\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\NARESH AS\Desktop\Agile_CourseProject\src\mai
n\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\NARESH AS\Desktop\Agile_CourseProject\src\tes
t\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
```

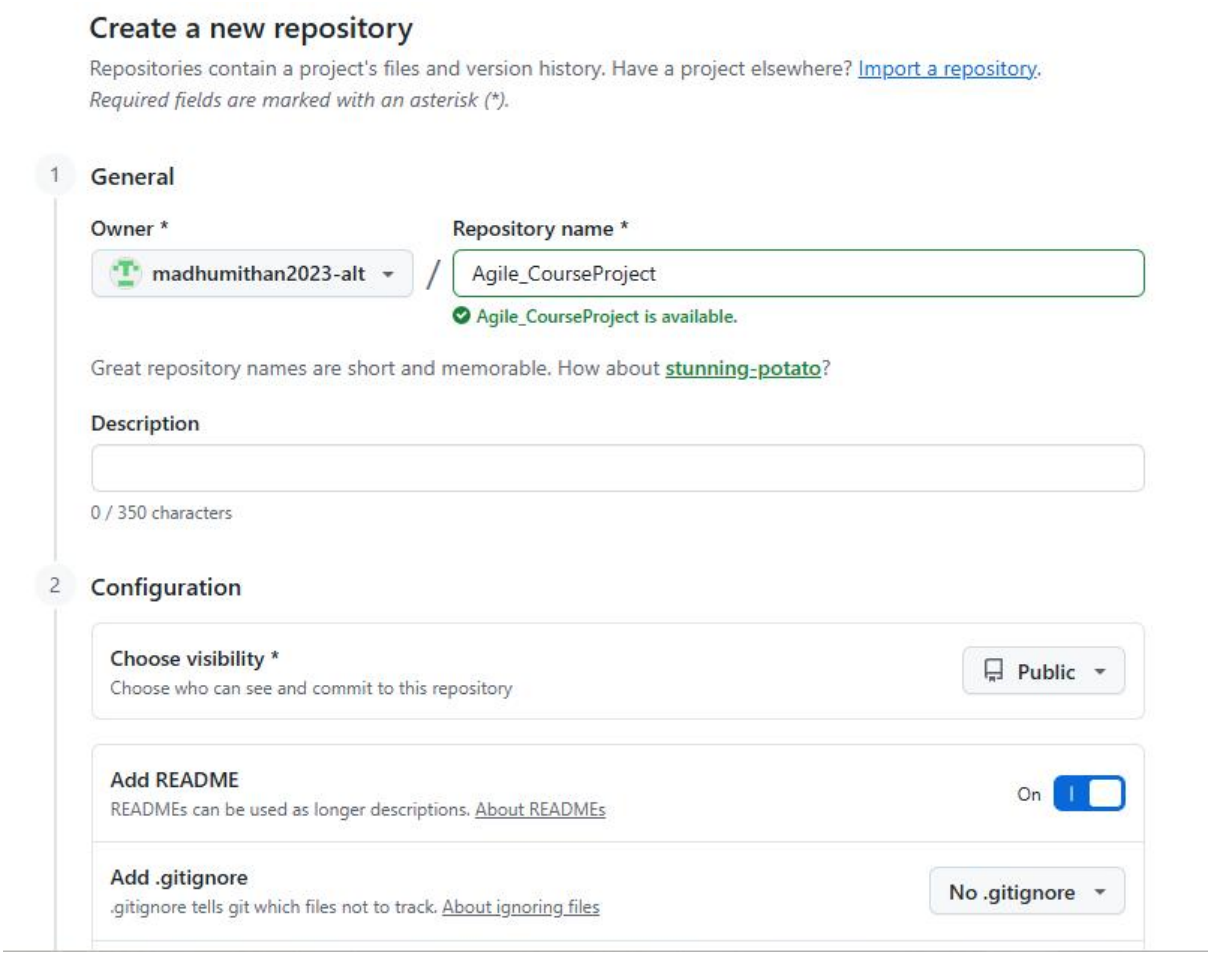
```
[INFO] Running AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.067 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.007 s
[INFO] Finished at: 2026-04-12T18:13:18+05:30
[INFO] ------------------------------------------------------------------------
```

mvn clean package

```
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\NARESH AS\Desktop\Agile_CourseProject\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\NARESH AS\Desktop\Agile_CourseProject\src\mai
n\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\NARESH AS\Desktop\Agile_CourseProject\src\tes
t\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
```

```
[INFO] Running AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.063 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\NARESH AS\Desktop\Agile_CourseProject\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.863 s
[INFO] Finished at: 2026-04-12T18:13:48+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject>
```

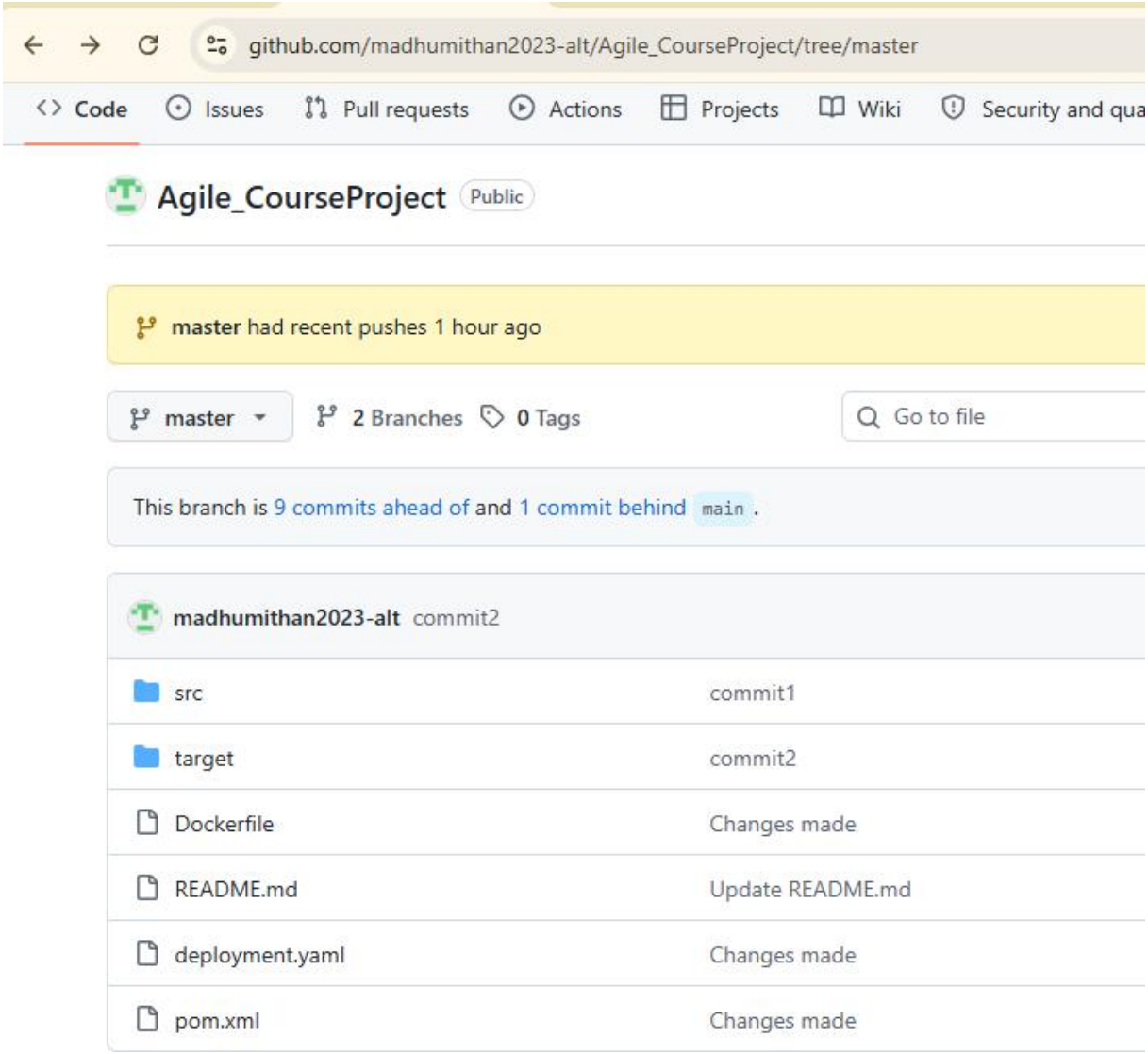
Create a new git repository



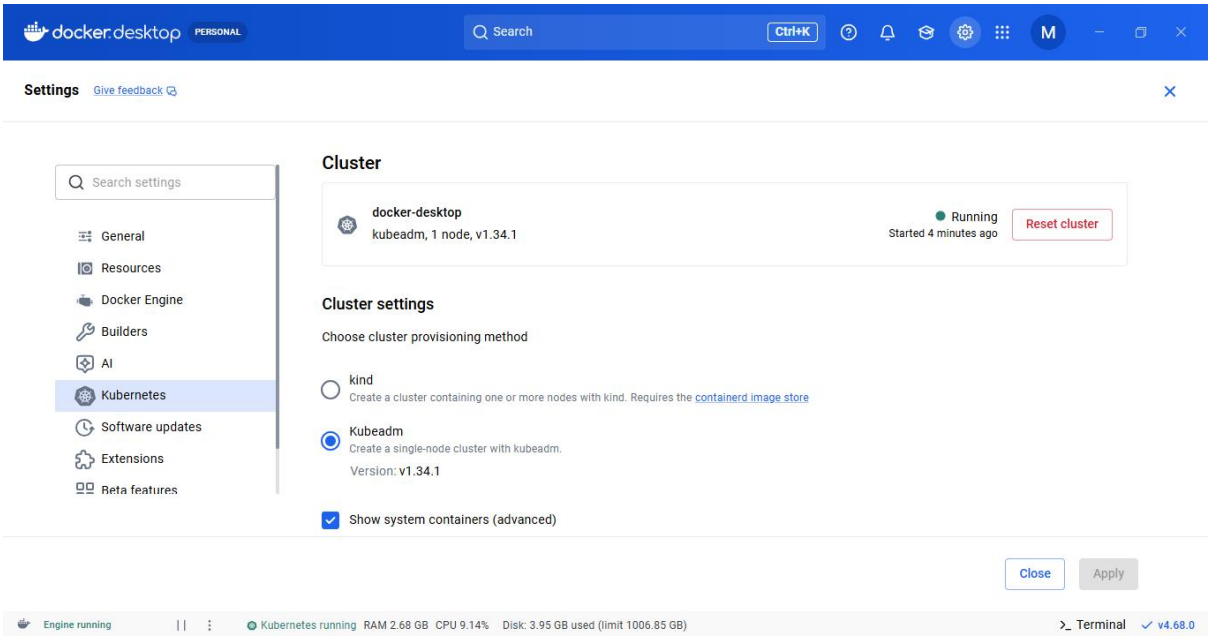
Pushing folder into git repository

```
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git init
Reinitialized existing Git repository in C:/Users/NARESH AS/Desktop/Agile_CourseProject/.git/
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFil
es.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles
.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/cr
eatedFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/in
putFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-AppTest.xml', LF will be replaced by CRLF the n
ext time Git touches it
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git add .
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git commit -m "commit2"
[master 18783c4] commit2
 2 files changed, 5 insertions(+), 5 deletions(-)
```

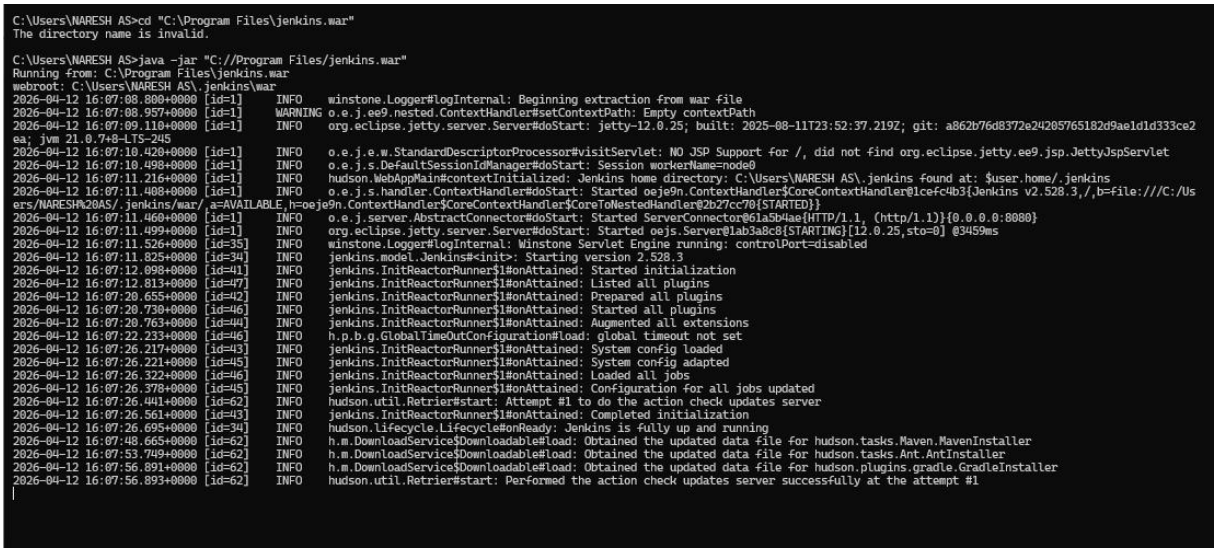
```
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git remote add origin https://github.com/madhumithan2023-a
lt/Agile_CourseProject
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git remote -v
origin  https://github.com/madhumithan2023-alt/Agile_CourseProject (fetch)
origin  https://github.com/madhumithan2023-alt/Agile_CourseProject (push)
PS C:\Users\NARESH AS\Desktop\Agile_CourseProject> git push -u origin master
Enumerating objects: 130, done.
Counting objects: 100% (130/130), done.
Delta compression using up to 4 threads
Compressing objects: 100% (100/100), done.
Writing objects: 100% (130/130), 29.45 KiB | 233.00 KiB/s, done.
Total 130 (delta 30), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (30/30), done.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/madhumithan2023-alt/Agile_CourseProject/pull/new/master
remote:
To https://github.com/madhumithan2023-alt/Agile_CourseProject
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```



Enable Kubernetes in Docker Desktop



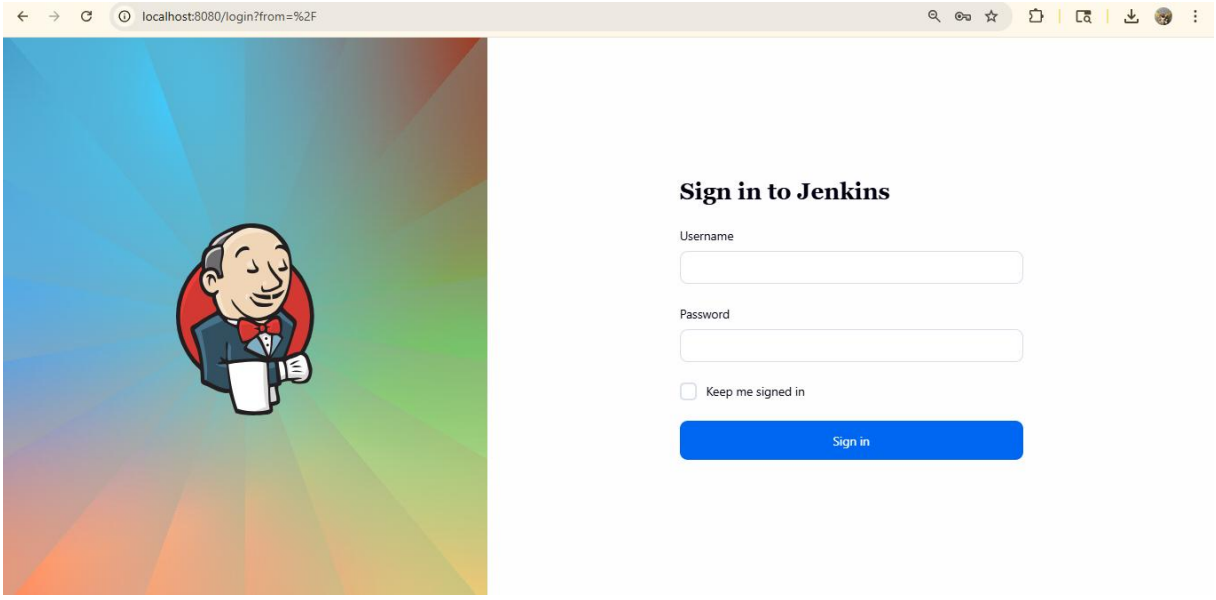
Run java -jar “C://Program Files/jenkins.war” in cmd



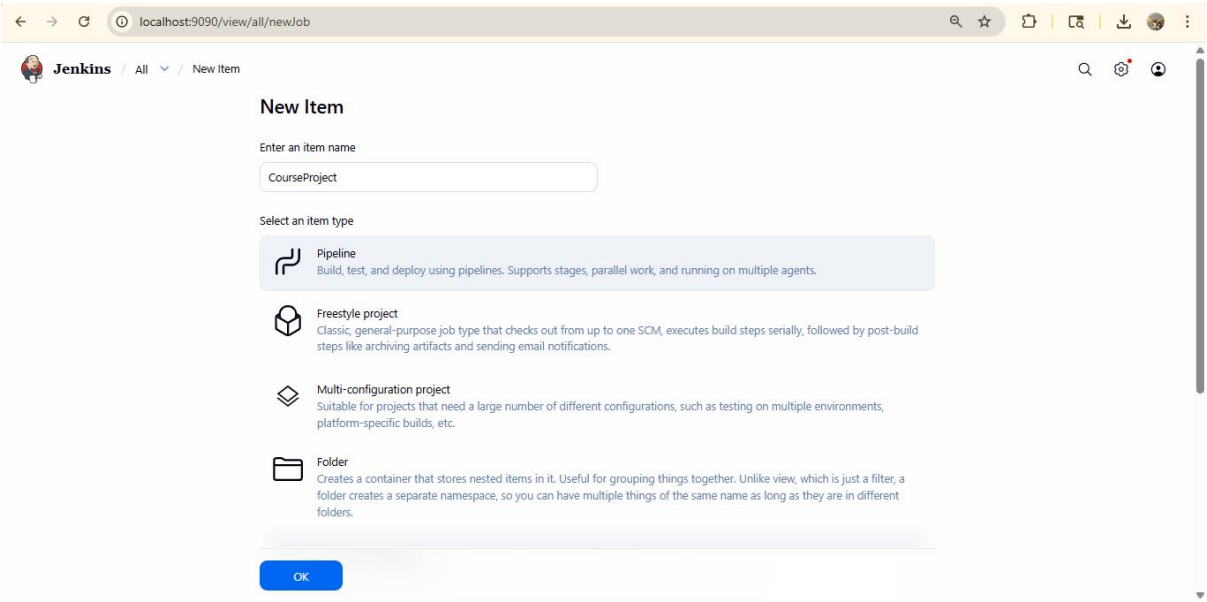
```
C:\Users\NARESH AS>cd "C:\Program Files\jenkins.war"
The directory name is invalid.

C:\Users\NARESH AS>java -jar "C://Program Files/jenkins.war"
Running from: C:\Program Files\jenkins.war
webroot: C:\Users\NARESH AS\.jenkins\war
2026-04-12 16:07:08.800+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-04-12 16:07:08.957+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-04-12 16:07:09.110+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2
ea; jvm 21.0.7+8-LTS-245
2026-04-12 16:07:10.420+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-04-12 16:07:10.498+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-04-12 16:07:11.216+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\NARESH AS\.jenkins found at: $user.home/.jenkins
2026-04-12 16:07:11.408+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@1cefc4b3{Jenkins v2.528.3,/,b=file:///C:/Us
ers/NARESH%20AS/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@2b27cc70{STARTED}}
2026-04-12 16:07:11.460+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started ServerConnector@61a5b4ae{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-04-12 16:07:11.499+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@1ab3a8c8{STARTING}[12.0.25,sto=0] @3459ms
2026-04-12 16:07:11.526+0000 [id=35]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-04-12 16:07:11.825+0000 [id=34]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.528.3
2026-04-12 16:07:12.098+0000 [id=41]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-04-12 16:07:12.813+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-04-12 16:07:20.655+0000 [id=42]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-04-12 16:07:20.730+0000 [id=46]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-04-12 16:07:20.763+0000 [id=44]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-04-12 16:07:22.233+0000 [id=46]    INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-04-12 16:07:26.217+0000 [id=43]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-04-12 16:07:26.221+0000 [id=45]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-04-12 16:07:26.322+0000 [id=46]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-04-12 16:07:26.378+0000 [id=45]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-04-12 16:07:26.441+0000 [id=62]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-04-12 16:07:26.561+0000 [id=43]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-04-12 16:07:26.695+0000 [id=34]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-04-12 16:07:48.665+0000 [id=62]    INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-04-12 16:07:53.749+0000 [id=62]    INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Ant.AntInstaller
2026-04-12 16:07:56.891+0000 [id=62]    INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.plugins.gradle.GradleInstaller
2026-04-12 16:07:56.893+0000 [id=62]    INFO    hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
```

Login jenkins using “localhost:8080”



Create new Item and select type as Pipeline

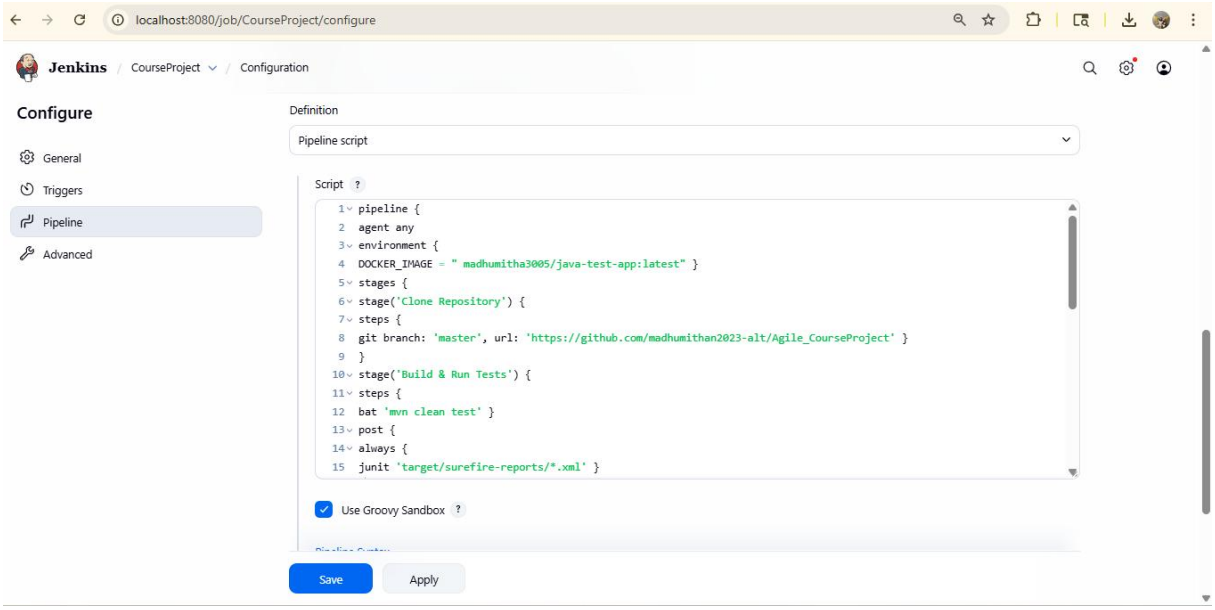


Pipeline script

```
pipeline {
agent any
environment {
DOCKER_IMAGE = " madhumitha3005/java-test-app:latest" }
stages {
stage('Clone Repository') {
steps {
git branch: 'master', url: 'https://github.com/madhumithan2023-
alt/Agile_CourseProject' }
}
stage('Build & Run Tests') {
steps {
bat 'mvn clean test' }
post {
always {
junit 'target/surefire-reports/*.xml' }
}
}

stage('Package Application') {
steps {
bat 'mvn package -DskipTests' }
}
stage('Build Docker Image') {
steps {
bat "docker build -t %DOCKER_IMAGE% ." }
}
stage('Push to DockerHub') {
steps {
withCredentials([usernamePassword(
credentialsId: 'dockerhub-creds', usernameVariable: 'DOCKER_USER',
passwordVariable: 'DOCKER_PASS' )]) {
bat """ docker login -u %DOCKER_USER% -p %DOCKER_PASS%
docker push %DOCKER_IMAGE%
```

```
""" }
}
}
stage('Deploy to Kubernetes') {
steps {
bat 'kubectl apply -f deployment.yaml' }
}
}
}
```



OUTPUT:

Apply, Save and Build now

kubectl get svc

```
C:\WINDOWS\system32\cmd.

Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NARESH AS>kubectl get svc
NAME           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-service   NodePort    10.103.74.184   <none>        80:30008/TCP   7m
kubernetes     ClusterIP   10.96.0.1       <none>        443/TCP        29m
```